Stage 1 — Data Assembly

- Import heterogeneous sources (FAO, World Bank, USDA, ...)
- Compute year-on-year relative changes
- Assemble candidate datasets by geographic resolution (country-level vs regional aggregates)
- 12 monthly price targets × 12 forecast horizons

Stage 2 — Screening Predictors

- 5 screening models: RF, GBM, CART, LM, GAM
- Leave-one-out cross-validation (from year 2000+)
- Extract variable importance per model, normalise to [0, 1]
- Error-weighted ensemble ranking across models
- Keep top 15 / 19 / 23 predictors -> screened datasets

Stage 3 — Generating Forecasts

- 7 ML models + 3 time-series models (ARIMAX, VAR, TBATS)
- Expanding-window CV (training from obs. 45+)
- 3 screened dataset sizes, auto-selected by cum. error
- 7 evaluation metrics (MAE, RMSE, RA, MAPE, R^2, CP, QL)
- Configurable metric -> best model per month × horizon

Stage 4 — Explaining Forecasts

- Retrain best model per CV fold (expanding window + LOO-CV)
- Shapley values (kernel SHAP via iml package)
- LIME local surrogates, permutation importance (with 90% CI)
- Friedman's H-statistic for feature interactions
- Intensity / Impact metrics -> 4 communication levels (L1–L4)